Specialization and Coordination in a Multi-Agent Technical Interview System: Trajectory Dynamics and Architectural Trade-offs

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Abstract. Educational and conversational systems that combine automated evaluation, large language models (LLMs), and reinforcement-learning controllers may be described as multi-agent systems, even when a single monolithic controller invokes components procedurally. While recent work compares agent configurations by the resulting task-artifact quality or aggregate benchmark performance, how agent specialization affects decision and trajectory behavior inside an already-deployed adaptive decision system remains less directly examined. We investigate whether decomposing interview decision-making into specialized, stateful agents with a coordination layer yields measurable behavioral, trajectory, or decision-making differences over a single-controller baseline. We extend an orchestrated technical-interview preparation system with two bounded-authority agents: a Strategy Agent managing a persistent weakness ledger across turns (proposing difficulty transitions and follow-up probes) and a Coaching Agent managing a coaching history ledger (proposing feedback depth, hints, and instructional emphasis), coordinated via unidirectional peer messaging and threshold-gated arbitration. Evaluating across 160 deterministic replay sessions (a fixed set of 64 technical-interview replay instances across 20 question orders; no live LLM was invoked), we find that adding the Strategy/Coaching agent layer alters difficulty trajectories in all 160 sessions relative to the frozen single-policy Proximal Policy Optimization (PPO) baseline, an effect the ablations associate with the Strategy proposal and arbitration path, with M_1 exhibiting fewer level changes ($15.12 \rightarrow 12.45$; cluster-bootstrap 95% CI $[-4.50, -0.24]$) and reversals ($11.74 \rightarrow 9.87$; cluster CI $[-3.12, -0.27]$) than the baseline, while reducing agreement with the baseline PPO action ($0.626 \rightarrow 0.400$; cluster-bootstrap 95% CI $[-0.284, -0.170]$) and introducing $+0.59$ probe turns per session, with the additional probe turns occurring only in replay cluster $k = 1$. However, in an open-loop replay harness, the split and unified implementations produced identical per-turn decisions ($\mathbb{P}(\text{agree}) = 1.0$, 1,280/1,280 turns). In the matched split-versus-unified comparison, specialization did not alter observed decision behavior; instead, it established separate state ledgers alongside small measured dispatch and serialization overheads (0.0440 vs 0.0401 ms; 0.0092 vs 0.0038 ms); relative to B_0, the multi-agent layer as a whole added $+1.04$ ms median per-turn latency. Threshold sensitivity experiments (τ from 0.0 to 0.9) indicate that trajectory behavior is sensitive to the confidence threshold in this replay configuration. We make no claims of human candidate learning or live model performance.

Keywords: multi-agent coordination · agent specialization · decision-making · trajectory dynamics · interview practice · arbitration

1 Introduction

Interactive educational systems that combine automated scoring, language models, and reinforcement-learning (RL) policies can be framed as multi-agent systems in the manner of recent LLM agent frameworks [8, 17], even when a single monolithic controller executes every component in a fixed procedural pipeline. Whether the multi-agent label reflects a genuine scientific distinction depends on how authority and state are distributed [16]. If components operate as specialized autonomous agents, questions arise regarding coordination mechanisms, trajectory dynamics, and architectural specialization: how do specialized agents coordinate their recommendations, how does their interaction affect session trajectories, and does partitioning responsibilities across specialized agents produce measurable behavioral differences or operational trade-offs compared to a unified controller? Recent comparative studies report task-dependent differences in software-engineering artifact validity across agent configurations [9] and a capability-dependent benefit of multi-agent over single-agent architectures on benchmark tasks [10]. These studies emphasize task-artifact quality or aggregate benchmark performance rather than the decision- and trajectory-level behavior examined here within an existing adaptive, reinforcement-learning-gated decision system.

We investigate these questions within an orchestrated spoken technical-interview preparation system whose deployed architecture we audit rather than assume. In its baseline configuration, the system executes an automated interview workflow driven by a single learned policy (a frozen Proximal Policy Optimization (PPO) controller [12, 14]) operating over a 6-dimensional state vector without concept memory. Other decision logic is governed by static threshold rules in the central hub. We extend this system by introducing two specialized, stateful agents behind a feature flag, leaving the underlying evaluator, frozen PPO policy, and candidate contract intact:

1. **Strategy Agent:** Maintains a private weakness ledger across turns, tracking persistent conceptual deficiencies and score trajectories to propose dynamic difficulty transitions and proactive follow-up probes.
2. **Coaching Agent:** Maintains a private coaching history ledger, tracking previously delivered hints and examples to propose feedback depth, instructional emphasis, and targeted remediation tags.

The agents possess strictly bounded authority: they only *propose*. A deterministic coordinator validates all proposals against formal schemas and authority matrices, an arbitrator resolves competing recommendations against deterministic guardrails and the PPO policy, and the central hub remains the sole system of record. Crucially, the Strategy Agent communicates pedagogical intent to the Coaching Agent via unidirectional peer messaging, conditioning feedback presentation without coupling private state ledgers.

This architecture enables a rigorous empirical evaluation of multi-agent coordination and specialization. Rather than evaluating generic agent reliability or treating multi-agent design as an axiomatic virtue, we formulate three precise research questions:

RQ1 (Coordination and intervention dynamics): How does adding the stateful Strategy/Coaching agent layer affect the decisions and session trajectories of the interview preparation system, compared with the frozen single-policy baseline (B_0)?

RQ2 (Empirical effects of agent specialization): Does separating Strategy and Coaching into distinct specialized agents yield measurable decision divergence or operational trade-offs compared to a unified technical coaching agent operating on equivalent information and tasks?

RQ3 (Coordination mechanisms): How do the implemented coordination mechanisms—stateful memory ledgers, peer messaging, confidence threshold gating, and PPO interaction—affect observed replay decision behavior and trajectory dynamics?

Summary of Findings and Contributions.

- *Coordination Dynamics (RQ1):* In deterministic replay across 160 sessions (operating on a fixed set of 64 technical-interview replay instances across 20 question permutation orders), the Strategy/Coaching agent layer alters difficulty trajectories in all 160 sessions relative to the frozen baseline. The M_1 trajectories show fewer level changes and reversals than the baseline, with mean level changes decreasing from 15.12 to 12.45 per session (-2.67, cluster-bootstrap 95% CI $[-4.50,-0.24]$) and mean reversals from 11.74 to 9.87 (-1.88, cluster CI $[-3.12,-0.27]$). Agreement with the uncoordinated PPO action decreases from 0.626 to 0.400 (mean difference -0.226, cluster CI $[-0.284,-0.170]$). Proactive probes introduce an average of $+0.59$ additional turns per session (cluster CI $[+0.00,+1.76]$). However, a cluster sensitivity analysis reveals that the additional probe turns occur only in replay cluster $k = 1$; omitting cluster 1 yields $\Delta = +0.00$ extra turns.
- *Limits of Specialization (RQ2):* Under the tested deterministic replay configuration, the split and unified architectures produced identical per-turn decisions, including difficulty actions, across all 160 sessions (1,280 turns; $\mathbb{P}(\text{agree}) = 1.0$), showing that, under the matched replay conditions, architectural decomposition did not produce different observed decision behavior. Ablating the Coaching Agent (M_3) reproduces the full two-agent system's difficulty trajectory in 160 of 160 sessions, as difficulty adaptation is structurally independent of coaching state. Relative to B_0, the multi-agent layer adds $+1.04$ ms median turn latency (47.7 agent calls per session); split versus unified was measured only at the dispatch (0.0440 vs 0.0401 ms) and serialization (0.0092 vs 0.0038 ms) level. Under the tested configuration, specialization establishes separate state ledgers and independent decision records without altering observed decision behavior, alongside measured operational overhead.

– *Mechanism Decomposition (RQ3):* Controlled ablations and a multi-level threshold sweep (τ from 0.0 to 0.9) indicate that trajectory behavior changes only when τ crosses the discrete realized confidence values (0.45, 0.70, 0.80) in the tested replay configuration. Setting $\tau = 0.0$ produced the same level-change (2.225), reversal (1.000), and PPO-agreement (0.014) values observed in M_4, despite retaining the other tested validation and guardrail conditions. Peer messaging (M_6) is designed to condition coaching depth and tone and leaves difficulty trajectories unchanged. Disabling PPO (M_7) alters trajectories in only 4 of 160 sessions (2.5%), contradicting the registered prediction of $\geq 50\%$ divergence and showing high agreement between the Strategy Agent heuristic and the baseline rule fallback.

What Kind of Contribution This Is. The specialized agents, typed contracts, and authority matrix constitute *architecture*; the peer messaging and threshold gating constitute *coordination mechanisms*; the replay ablations and cluster-bootstrap evaluations provide *empirical evidence on coordination and the limits of specialization.*

Explicit Disclaimers. All replay experiments execute deterministically with recorded evaluator outputs (`EVALUATOR_MOCK_MODE=1`); no live large language model (LLM) was invoked during replay. We make **no claims** regarding candidate learning, pedagogical efficacy, interview placement success, or LLM hallucination rates. The lower reversal count reflects the observed trajectory metrics under the tested replay configuration, not proven pedagogical efficacy.

2 Related Work

Multi-Agent Coordination and Specialization. The classic characterization of agency by Wooldridge and Jennings defines agents through autonomy, reactivity, pro-activeness, and social ability [16]. In modern LLM-based systems, multi-agent frameworks predominantly follow two coordination paradigms: conversational emergent interaction, exemplified by AutoGen [17], and structured role-based workflows with standardized operating procedures, exemplified by MetaGPT [8]. Although these frameworks are versatile, a recent empirical study identifies recurring failure modes in multi-agent LLM systems, including specification issues, inter-agent misalignment and task-verification failures [3]. This motivates an empirical question not resolved by architectural descriptions alone: whether specialization produces measurable decision differences or operational trade-offs relative to a unified controller operating with equivalent information. **Agent Configuration and Architectural Comparison.** Kim [9] compares Task-Specialized, Phase-Specialized, and Process-Generalist agent configurations across software-development tasks, evaluating output-artifact validity and finding that no single configuration dominates across tasks. Separately, Kim et al. [10] compare coordination structures and base-model capability across benchmark tasks, finding that the benefit of multi-agent coordination over a single agent diminishes, and can reverse, beyond a capability-saturation threshold. These studies illustrate that the effect of architectural and configuration choices can depend on task and

model capability, and that controlled comparisons are useful for isolating those effects. They emphasize task-artifact quality or aggregate benchmark performance rather than the decision- and trajectory-level behavior examined here within an existing adaptive, reinforcement-learning-gated decision system. Those studies involve LLM-driven agents; the agents studied here are deterministic and rule- and ledger-based. The proposed system examines this distinct evaluation target under an explicit, code-verified authority and information topology (Strategy may alter difficulty; Coaching may not; messaging is unidirectional), letting coordination (a stateful agent layer vs. a non-agentic baseline) and specialization (split vs. unified roles) be varied and measured as separate architectural variables within the same system.

Adaptive Mock Interview and Tutoring Systems. AI-assisted interview systems have emerged to provide automated feedback and dynamic evaluation for technical candidates. Systems such as Zara [18] utilize generative LLMs to synthesize candidate feedback, while empirical user studies investigate student confidence, readiness, and trust in automated code evaluation [7, 15]. However, generative LLM evaluators are known to exhibit position bias and verbosity bias [19]. To ensure evaluation integrity, our target system enforces strict authority boundaries derived from the principle of least privilege [13]: scoring is computed by a deterministic, cross-encoder evaluator outside the language model, while agents operate solely on validated observations.

Reinforcement Learning, Shielding, and Guardrailed Arbitration. Dynamic difficulty adaptation can be framed as a reinforcement learning problem. Single-agent RL approaches, such as Proximal Policy Optimization (PPO) [14] implemented via Stable-Baselines3 [12], optimize policy returns over state trajectories. However, unconstrained RL policies risk erratic behavior during out-of-distribution candidate interactions [6]. Safe reinforcement learning surveys [6] examine external constraint mechanisms, such as shielding [1], that intervene on learned actions when specific constraints are violated. Our arbitration architecture implements deterministic guardrails and threshold gating where rule-based, evidenced agent proposals compete with the frozen PPO action while remaining bounded by verified transitions. This architecture represents centrally orchestrated role specialization rather than Multi-Agent Reinforcement Learning (MARL) [2], as the PPO policy is frozen and the agents do not update policy weights dynamically.

Methodological Foundations and Statistical Clustered Inference. For statistical inference over the 8 underlying replay clusters, we employ clustered bootstrapping methodologies established by Field and Welsh [5], preventing artificially narrow confidence intervals that arise from treating permuted question sequences as independent evaluation sessions. To evaluate architectural boundaries, we draw upon interface-isolation practices from production software engineering [11], ensuring that agent interfaces maintain decoupled execution scopes.

Table 1. Authority matrix, generated from the enforcing code (`agents/multiagent/authority.py`). Neither agent has state-write authority; the hub commits alone.

Agent	May propose	Private state	Cannot	On invalid output
Strategy	difficulty_action (Easier/Same/Harder); probe request (concept, kind); strategy mode	weakness ledger (per-concept miss counts, probe history, trend)	write or overwrite any score; set a difficulty level directly; terminate the session; edit history; override hard constraints or guardrail overrides; call another agent	invalid / stale / low-confidence / timed-out proposal is dropped; PPO proposal stands
Coaching	feedback depth; hint / example flags; concepts to emphasise; remediation tag; tone	coaching history (hints/examples given, repeated misconceptions)	alter evaluator results or scores; cite concepts absent from the evaluator evidence; request difficulty changes; request follow-ups; terminate the session	invalid plan is dropped; the existing template feedback is used unchanged

3 The Existing System and What Was Added

3.1 Existing workflow (preserved)

A browser client sends a spoken answer to a backend, which transcribes it and calls an *InterviewOrchestrator* (the hub; Fig. 1). The hub owns one session-state dictionary and one lock and runs a fixed per-turn pipeline: (1) a learned *evaluator* (sentence-embedding similarity S_1, a concept index S_2 and a fine-tuned cross-encoder R; $0.15S_1+0.35S_2^{\text{eff}}+0.50R$ with a cap when a mandatory concept is missed) returns a score and concept evidence; (2) a strict hand-off contract treats a non-finite or out-of-range score as an evaluator failure and fails closed; (3) feedback is produced by a template or an optional local LLM; (4) the hub commits the turn; (5) a baseline schedule and then a frozen PPO policy [12, 14] propose an easier/same/harder action, followed by a rule layer of six guardrails; (6) a threshold rule decides whether a follow-up question is injected (at most two in a row). Before this work only the PPO policy was a reactive learned decision-maker; the LLM never chose an action. All of this is unchanged when the multi-agent flag is off, and a test shows that with the flag on but no agent contributing the trajectory is identical (Sect. 6).

3.2 Agents

Table 1 lists what each agent may propose and may not do. Applying the principle of least privilege [13], neither agent possesses state-write privileges.

Both agents implement a single method `propose(observation, peer)`, receive an immutable, validated `TurnObservation` (score, evidence lists, difficulty, score history, progress, follow-up counters; counters carry the value the hub will commit), and return a typed proposal. They are rule- and ledger-based, not LLM-driven; they are the smallest design that has private persistent state, pro-active proposals and a communication channel, which are the properties of weak agency [16] that the existing roles lacked. We make no claim of autonomy beyond that.

Strategy Agent. Objective: maintain configured difficulty adaptation while tracking persistent weaknesses within the follow-up budget. Private state: per concept, the distinct turns in which the evaluator listed it as missing or incorrect, which concepts were later covered, probes already requested, and an exponential moving average (EMA) of scores. Rules, first match wins: *recover* (current score < 0.35 and mean of the last two < 0.5, or a weakness that survived its probes while the score stays < 0.5) $\rightarrow$ Easier; *remediate* (a concept missed on ≥ 2 distinct turns and unresolved) $\rightarrow$ Same plus a probe request for that concept; *stretch* (score ≥ 0.8, average ≥ 0.7, last three ≥ 0.6, no open weakness) $\rightarrow$ Harder; otherwise *steady*. Non-*steady* proposals use confidence $\min(0.95, 0.5{+}0.1n)$ for n cited evidence items (a documented heuristic rather than a calibrated probability), while *steady* proposals are assigned 0.45 (below the arbitration threshold $\tau = 0.6$, so the PPO action stands unless the ledger supports a change).

Coaching Agent. Objective: choose how to coach after an evaluated answer. Private state: concept miss counts and the hints and examples already given. Decisions: up to two concepts to emphasize (incorrect claims first, then repeatedly missed, then evaluator order); a hint if concepts are missing, the score is < 0.6 and fewer than two hints were given for the lead concept; an example when the lead concept was missed before and no example was given yet; depth (*detailed* if score < 0.4 or a repeated miss, *brief* if ≥ 0.85 with nothing missing); tone; a remediation tag. A Strategy proposal delivered as a peer message modifies the plan (*recover*: detailed and encouraging; *stretch*: brief; *remediate*: remediation tag on the probed concept). A deterministic renderer applies the plan to the existing feedback dictionary: it adds presentation fields and may prepend a priority tip to, or truncate, the existing improvement and communication-tip lists, and it raises if any protected evaluator field (score, grade, concept and claim lists, transcript, decision source) would change.

3.3 Communication and state ownership

Every observation, proposal and decision is a typed object (`pydantic`, unknown fields rejected, NaN and out-of-range values rejected). The coordinator records each call as an envelope with session, turn, sender, recipient, state version and provenance, and every accept/reject/modify outcome as a decision record with a reason code. Optimistic concurrency uses a *state version* that counts committed turns: a proposal whose version, turn or session differs from the observation, or from the hub's current version, is stale and dropped; a second answer from the

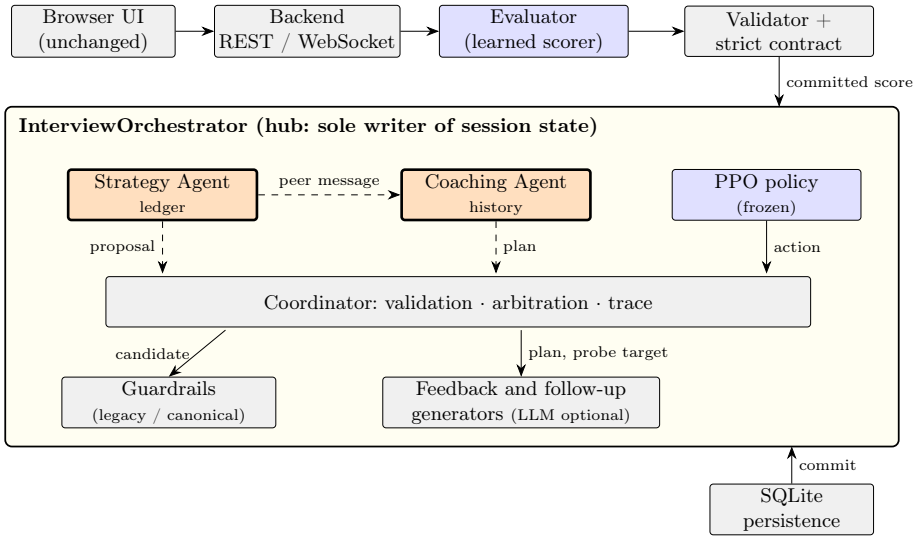


Fig. 1. Implemented architecture. Shaded orange: the two added agents; blue: learned components; grey: deterministic infrastructure. Dashed arrows are typed proposals; agents never write state.

same agent for the same version is a duplicate and dropped. The hub builds observations, commits state, increments the version, and persists agent memory in its session snapshot so that a recovered session resumes with the agents' ledgers.

4 Orchestration, Arbitration and Validation

Figure 2 shows one turn. **Difficulty precedence** (deterministic code, evaluated by ablation rather than assumed optimal): (1) hard constraints (infrastructure failure forces Same; levels stay in $[1, 5]$); (2) the guardrail layer, applied last to whichever candidate was selected, wins whenever a rule fires (G3 and G6 hold or raise difficulty, so this layer does not act solely in a downward direction); (3) a validated Strategy proposal with confidence $\geq \tau$ and cited evidence supersedes the PPO action, recorded as a *conflict*; (4) the PPO action, or the labeled non-RL heuristic when PPO is unavailable. A proposal that cannot take effect at the bounds is recorded as *modified*. **Probe precedence:** the hard cap of two consecutive follow-ups, then a Strategy probe whose concept appeared in the evaluator's evidence earlier in this session and whose confidence is $\geq \tau$ (it may add a probe the thresholds would not trigger, or retarget the default one), then the hub's original threshold rule. The retargeted concept is passed to the template follow-up generator; the LLM prompt is not changed.

PPO integration. PPO keeps its role and its 6-D state and 3-action definitions; the checkpoint is frozen. Its action is one input to arbitration. The agents never see the PPO action before they propose, and the paper makes no claim that the agents improve PPO. Calling this multi-agent RL would be wrong: one policy is learned, and the agents are not trained.

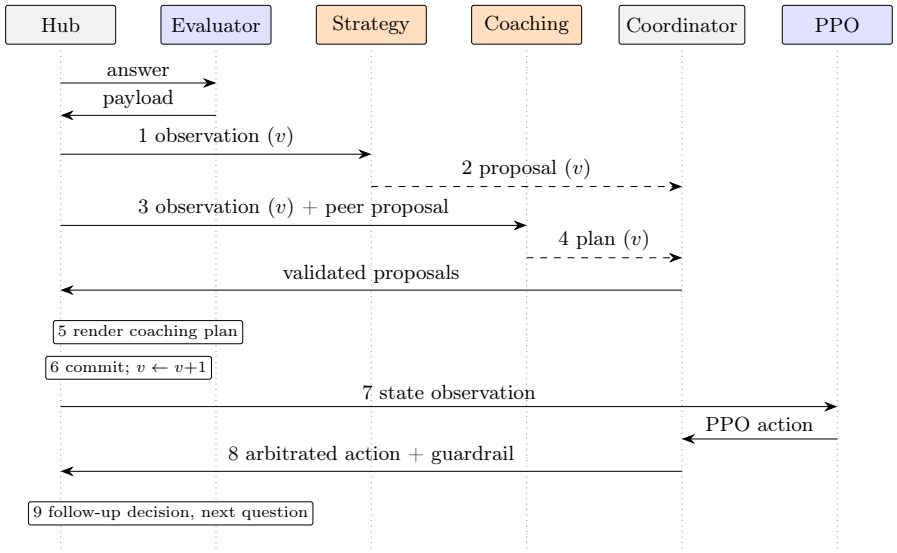


Fig. 2. One interview turn as implemented (verbal path). Steps 1–4 happen before the hub commits; arbitration (8) follows the PPO suggestion. On the code-submission path only the Strategy proposal is collected.

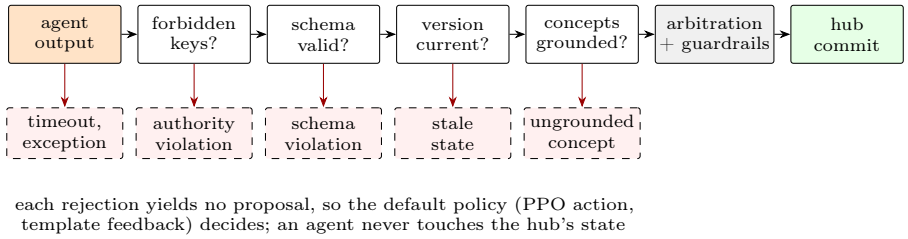


Fig. 3. Validation pipeline every proposal passes before arbitration. Each rejection yields no proposal; the default policy decides.

Guardrails. The deployed hub uses an inline copy of the rule layer that differs from the module used by the training environment. A configuration key selects `legacy_inline` (default; deployed behavior) or `canonical`; both are reported separately. On a synthetic grid (39,930 points, hesitation $hes = 1 - conf$, with $conf$ the acoustic confidence) the two disagree on 4,728 points (11.8%) in G2, G6 and the no-override case; this is a property of the grid, not a deployment frequency, and a regression test pins it.

Proposal Validation. Before any proposal reaches arbitration, the coordinator executes four sequential checks (Fig. 3): (1) strict Pydantic schema validation (`extra=forbid`, NaN and out-of-range rejection); (2) recursive scanning against the 18 forbidden state keys; (3) optimistic concurrency verification matching the hub's current committed turn version; and (4) concept evidence grounding. If an agent raises an unhandled exception or exceeds the configured timeout (150 ms), its proposal is dropped and the hub proceeds with the default policy.

5 Experimental Methodology

5.1 Evaluation Environment and Replay Dataset

To evaluate coordination and specialization without stochastic variance from live network calls, all experiments run via deterministic replay through the real orchestrator hub with `EVALUATOR_MOCK_MODE=1`.

Replay Stimuli and Dataset Structure. All experiments operate on a fixed set of 64 technical-interview replay instances across eight underlying replay clusters prepared for the proposed system, combining constructed (authored) answer payloads with team-generated/synthetic instances. In replay mode, the hub consumes these fixed evaluator inputs directly, driving the full state machine and coordinator. For follow-up probes, the replay reuses the recorded evaluator payload associated with the root concept rather than generating a new candidate response.

The 8-Cluster Structure. Replay sessions combine the 8 underlying replay clusters with 20 question orders (order seeds 0–19; seed 0 is the original order), yielding $N = 160$ sessions per experimental condition. These 160 sessions are **not independent**; they reuse the identical 8 replay clusters. We therefore explicitly model the 8 underlying replay clusters as the independent structural units in the cluster-aware analyses.

Evaluation Mode. No live generative LLM is invoked during replay; the LLM gateway is stubbed as down and deterministic template renderers handle text output, so trajectory differences are evaluated under fixed stimuli without live-model stochasticity.

5.2 Baseline and Ablation Matrix

To address RQ1 (coordination dynamics) and RQ3 (coordination mechanisms), we evaluate the baseline platform alongside controlled ablation conditions (Experiment XA) across the identical 160 replay sessions:

- **Frozen Baseline (B_0):** Single-controller workflow without agents, with PPO active and deployed inline guardrails.
- **PPO-Disabled Baseline (B_7):** Single-controller baseline with PPO unavailable, using deterministic rule fallback.
- **Canonical-Guardrail Baseline (B_{0c}):** Baseline evaluated with canonical `rl/guardrails.py` instead of inline rules.
- **Full Multi-Agent System (M_1):** Strategy and Coaching agents, peer messaging, and arbitration ($\tau = 0.6$).
- **Strategy Agent Ablated (M_2):** Strategy disabled while Coaching and the coordination layer remain active.
- **Coaching Agent Ablated (M_3):** Coaching disabled while Strategy remains active, testing trajectory independence.
- **Arbitration Ablated (M_4):** Raw Strategy proposals admitted without confidence-threshold, evidence-grounding, or step-bound checks.

- **Grounding Verification Ablated (M_5):** Coaching plans accepted without concept-evidence grounding checks.
- **Peer Communication Ablated (M_6):** Strategy-to-Coaching peer messaging disabled, isolating message exchange.
- **Authority Enforcement Ablated (M_8):** Score authority verification bypassed before proposal admission.
- **PPO-Disabled Multi-Agent (M_7):** Full multi-agent system with PPO disabled, falling back to rule heuristic (paired with B_7).
- **Canonical-Guardrail Multi-Agent (M_{1c}):** Full multi-agent system evaluated with canonical guardrails.

5.3 Framework for Evaluating Agent Specialization (RQ2)

To evaluate whether decomposing technical coaching into specialized agents produces measurable behavioral divergence or operational trade-offs compared to a unified controller (RQ2), we formulate a controlled comparative framework contrasting:

1. **Specialized Architecture ($M_{\textbf{split}}$):** The two-agent baseline (M_1) partitioning responsibilities across Strategy and Coaching with separate state ledgers and explicit peer messaging.
2. **Unified Architecture ($M_{\textbf{unified}}$):** A monolithic agent (`UnifiedCoachAgent`) maintaining a joint session ledger and synthesizing difficulty actions, follow-up probes, and coaching plans in a single dispatch.

Equivalence Conditions. To ensure an equivalent comparison, both configurations receive identical observation inputs, identical question sequences, identical heuristic decision thresholds, and identical arbitration rules. We evaluate four measured outcomes alongside one architectural property: (1) decision agreement rate ($\mathbb{P}(\text{Decision}_{\text{unified}} = \text{Decision}_{\text{split}})$); (2) difficulty-action agreement and (3) follow-up probe agreement, both components of (1); (4) per-turn dispatch and serialization latency; and (5) architectural state encapsulation (schema separation and separate ledger boundaries).

5.4 Threshold Decomposition (Experiment XC)

Because the unarbitrated condition (M_4) simultaneously removes the confidence threshold, evidence verification, and one-step limits, it cannot isolate which mechanism prevents oscillation collapse. We therefore execute Experiment XC, sweeping the confidence threshold $\tau \in \{0.0, 0.5, 0.6, 0.7, 0.8, 0.9\}$ across the same 160 replay sessions while holding all other guardrail boundaries and evidence checks active.

5.5 Statistical Methodology

For session-level point estimates and paired differences, we compute standard non-parametric percentile bootstrap intervals with $B = 10{,}000$ resamples. To

Table 2. Replay ablation summary across 160 sessions per condition (mean per session unless noted as counts). **compl./auth./trans.**: sessions completing, keeping committed scores authoritative, and preserving valid transitions. **chg./rev.**: mean level changes and reversals; **agr.**: fraction of decisions agreeing with the PPO action; **gfire**: guardrail overrides; **strat.**: decisions where the Strategy Agent was the final action source; **byp.**: sessions with guardrail bypass; **p50/p95**: pooled per-turn latency in ms; **calls**: agent calls per session.

Cond.	compl.	auth.	trans.	chg.	rev.	agr.	gfire	strat.	byp.	p50	p95	calls
B0	160	160	160	15.119	11.744	0.626	9.588	0.0	0	0.546	2.056	–
B7	160	160	160	2.675	1.256	–	3.831	0.0	0	0.463	1.059	–
B0c	160	160	160	15.119	11.744	0.626	9.588	0.0	0	0.553	2.004	–
M1	160	160	160	12.45	9.869	0.4	6.838	6.669	0	1.589	4.411	47.68
M2	160	160	160	15.119	11.744	0.626	9.588	0.0	0	1.218	3.513	23.25
M3	160	160	160	12.45	9.869	0.4	6.838	6.669	0	1.118	3.189	23.84
M4	160	160	160	2.225	1.0	0.014	0.0	19.619	3	1.573	4.357	47.68
M5	160	160	160	12.45	9.869	0.4	6.838	6.669	0	1.581	4.553	47.68
M6	160	160	160	12.45	9.869	0.4	6.838	6.669	0	1.592	4.46	47.68
M8	160	160	160	12.45	9.869	0.4	6.838	6.669	0	1.575	4.35	47.68
M7	160	160	160	2.706	1.312	–	3.812	9.037	0	1.502	2.968	47.68
M1c	160	160	160	12.45	9.869	0.4	6.838	6.669	0	1.573	4.534	47.68

account for the 8-cluster dependency, we implement a cluster bootstrap over the 8 underlying replay clusters following Field and Welsh [5] ($B = 10{,}000$), resampling entire replay clusters with replacement. In addition, we report a leave-one-cluster-out (LOCO) sensitivity analysis to verify whether observed effects depend on individual influential replay clusters. All reported bootstrap confidence intervals are descriptive measures of parameter dispersion across resamples; no formal null-hypothesis significance tests or p-values are evaluated.

6 Results

All reported results are derived from deterministic REPLAY of recorded evaluator outputs unless explicitly noted. All sessions within a condition are dependent across question orders due to the 8-cluster structure (Sect. 5).

6.1 Coordination and Intervention Dynamics (RQ1)

Execution Invariants and Baseline Transparency. Every evaluated condition completed all 160 sessions, maintained strict evaluator authority (committed scores equaled recorded evaluator output in 100% of sessions), and enforced valid difficulty bounds in $[1, 5]$ with transitions of at most one level per step (A1–A3, Table 4; exact Clopper–Pearson 95% lower bound for 160/160: 0.977 [4]; treating sessions as independent; descriptive). In the Strategy-ablated condition (M_2), difficulty, score, and question trajectories reproduced the baseline (B_0) identically in 160 of 160 sessions (A4), indicating that the coordination layer leaves these trajectories unchanged when Strategy is disabled and Coaching remains active.
Trajectory Divergence and Reduced Level Changes. Enabling the agent layer (M_1) altered the difficulty trajectory relative to the frozen baseline (B_0)

Table 3. Paired differences ($A - B$) over identical 160 replay sessions: fraction of sessions with divergent difficulty paths, paired mean differences with naive session-level 95% percentile-bootstrap intervals, and pooled median turn latency delta. Intervals are descriptive due to cluster reuse.

A−B	path≠	Δ level chg.	Δ reversals	Δ PPO agree	Δ p50 ms
M1−B0	1.0	−2.669 [−3.188,−2.131]	−1.875 [−2.275,−1.462]	−0.226 [−0.244,−0.208]	+1.043 [+1.000,+1.079]
M2−B0	0.0	0	0	0	+0.672 [+0.646,+0.698]
M3−M1	0.0	0	0	0	−0.471 [−0.511,−0.425]
M4−M1	1.0	−10.225 [−10.544,−9.900]	−8.869 [−9.275,−8.463]	−0.387 [−0.403,−0.370]	−0.016 [−0.065,+0.033]
M5−M1	0.0	0	0	0	−0.008 [−0.059,+0.037]
M6−M1	0.0	0	0	0	+0.004 [−0.030,+0.053]
M8−M1	0.0	0	0	0	−0.013 [−0.057,+0.032]
M7−B7	0.025	+0.031 [+0.000,+0.081]	+0.056 [+0.006,+0.125]		+1.039 [+0.996,+1.070]
B0c−B0	0.0	0	0	0	+0.008 [−0.019,+0.027]
M1c−M1	0.0	0	0	0	−0.016 [−0.055,+0.032]

in **all 160 replay sessions (100%)** (A7; Table 3). The M_1 trajectories show fewer level changes and reversals than B_0: mean level changes fell from 15.12 to 12.45 per session (point difference −2.67, naive session bootstrap 95% CI $[-3.19, -2.13]$), and mean reversals fell from 11.74 to 9.87 per session (−1.88, naive CI $[-2.28, -1.46]$). Agreement with the uncoordinated PPO action dropped from 0.626 to 0.400 (point difference −0.226, cluster CI [−0.284,−0.170]), with Strategy proposals winning pre-guardrail arbitration in 11.9 decisions per session and remaining the final unoverridden action source in 6.669 decisions after downstream guardrail evaluation (**strat.** in Table 2).

Cluster Dependence and Proactive Probing Sensitivity. Because the 160 sessions reuse 8 underlying replay clusters, treating sessions as independent underestimates parameter uncertainty. A post-hoc cluster bootstrap over the 8 replay clusters (Field and Welsh [5], $B = 10{,}000$) yields wider, defensible intervals: level changes −2.67 [−4.50,−0.24], reversals −1.88 [−3.12,−0.27], and PPO action agreement −0.226 [−0.284,−0.170]. The negative sign for level changes and reversals is shared across 7 of the 8 replay clusters.

Proactive probing added an average of +0.59 turns per session (naive CI $[+0.34, +0.85]$, cluster CI [+0.00,+1.76]). However, cluster-level decomposition reveals that this probe increase is present in **only 1 of the 8 replay clusters** ($k = 1$); in the remaining seven clusters, proactive probing added exactly 0.00 turns. A leave-one-cluster-out (LOCO) sensitivity analysis confirms that omitting cluster $k = 1$ shifts the level-change delta to −3.76 (CI $[-4.81, -2.78]$), whereas omitting any of the other seven clusters causes the 95% interval to include zero. Thus, while a directional reduction in level changes and reversals is observed across most replay clusters (7 of 8), the inferential magnitude is sensitive to cluster composition, and extra-turn gains cannot be generalized as an unconditional property.

Proposal Distribution. Across the 160 sessions, the Strategy Agent generated 3,814 proposals: 1,300 *remediate* proposals (confidence 0.80), 928 and 312 *recover* proposals (confidence 0.80 and 0.70), and 1,274 *steady* proposals (confidence 0.45). Notably, the Strategy Agent emitted zero *stretch* proposals (0), indicating that during replay the candidate trajectories never maintained the sustained high-performance criteria (3 scores ≥ 0.60, EMA ≥ 0.70) required to trigger proactive difficulty increases. **Coordination Overhead.** Introducing the multi-agent layer

Table 4. Registered predictions and empirical outcomes for Experiment XA.

ID	Prediction	Outcome
A1	completion 100% in every condition	Supported
A2	committed scores equal the recorded evaluator output in every session of every condition	Supported
A3	difficulty levels stay within [1,5], and every RL-phase decision moves by at most one level, in every session of every condition	Supported
A4	M2 (no Strategy Agent) reproduces B0 (difficulty, score and question paths) in every session	Supported
A5	M3, M5, M6, M8 reproduce M1's difficulty/score/question paths in every session	Supported
A6	M4 (no arbitration) has guardrail_bypass>0 in $\geq$1 session; M1 has 0 in all	Supported
A7	M1 differs from B0 in difficulty path in $\geq$50% of sessions	Supported
A8	median per-turn latency M1 − B0 $<$ 50 ms (pooled turns)	Supported
A9	with PPO unavailable, M7 differs from B7 in difficulty path in $\geq$50% of sessions	Contradicted
A10	canonical guardrails change the difficulty path in $\geq$1 session vs legacy (B0c vs B0)	Contradicted

Table 5. Experiment XC (REPLAY, 160 sessions per row, mean per session): only confidence threshold τ varies; evidence requirements and guardrails remain active. **chg./rev.**: mean level changes and reversals; **agr.**: agreement with PPO; **strat.**: decisions from Strategy Agent; **gfire**: guardrail overrides; **byp.**: sessions with guardrail bypass.

τ	chg.	rev.	agr.	strat.	gfire	byp.
0.0	2.225	1.000	0.014	16.075	3.544	0
0.5	12.450	9.869	0.400	6.669	6.838	0
0.6	12.450	9.869	0.400	6.669	6.838	0
0.7	12.450	9.869	0.400	6.669	6.838	0
0.8	12.450	9.819	0.404	6.594	6.863	0
0.9	15.119	11.744	0.626	0.000	9.588	0

increased pooled per-turn latency from 0.546 ms (B_0) to 1.589 ms (M_1) at the median (end-to-end orchestration overhead +1.04 ms, 95% CI $[+1.00, +1.08]$ ms) and from 2.056 ms to 4.411 ms at the 95th percentile, executing an average of 47.7 agent calls per session. This end-to-end orchestration overhead reflects the broader coordination path across the turn lifecycle, encompassing multi-agent invocation, peer observation passing, contract checking, and arbitration logic.

6.2 Coordination Mechanisms and Sensitivity (RQ3)

Decomposing Arbitration via Threshold Sensitivity (XC). In the unarbitrated condition (M_4), where all arbitration checks were simultaneously disabled, trajectory dynamics collapsed: level changes fell to 2.225 per session, PPO agreement plummeted to 0.014, and guardrail bypass occurred (Table 2). Experiment XC isolates the specific gating mechanism by varying only the confidence threshold τ from 0.0 to 0.9 while keeping the other validation, evidence, and guardrail conditions active (Table 5).

Setting $\tau = 0.0$ produced the same level-change (2.225), reversal (1.000), and PPO-agreement (0.014) values observed in M_4, despite retaining the other tested validation and guardrail conditions, with **zero guardrail bypass** in

any session. Because unevidenced *steady* proposals are assigned confidence 0.45, setting $\tau = 0.0$ allows uncalibrated default proposals to override PPO on most RL decisions (16.1 of 19.6 per session). This identifies the confidence threshold as a key sensitivity point in this replay configuration rather than absent guardrails or step bounds. Behavior remained flat across $\tau \in [0.5, 0.7]$ (12.450 level changes, 0.400 agreement; 0.404 agreement at $\tau = 0.8$) because realized evidenced confidences occur at discrete values (0.70 and 0.80). At $\tau = 0.9$, the threshold exceeds all realized heuristic confidences (max = 0.80), preventing Strategy proposals from superseding PPO and reverting trajectories to the baseline (B_0). All 5 registered XC predictions were supported.

Peer Information Exchange and Grounding Ablations (M_5, M_6). Ablating grounding verification (M_5) or the unidirectional peer message from Strategy to Coaching (M_6) resulted in difficulty trajectories identical to the full configuration (M_1) across all 160 sessions (A5). The peer message does not feed back into Strategy decision-making; its role is strictly to condition Coaching presentation (switching depth to *detailed* during *recover*, *brief* during *stretch*, and attaching remediation tags during *remediate*).

Contradicted Prediction A9 (PPO Fallback Interaction). Registered prediction A9 posited that disabling the learned PPO policy would alter $\geq 50\%$ of session paths. Empirically, disabling PPO (M_7) altered difficulty trajectories in only **4 of 160 sessions (2.5%)**. Inspection of orchestrator traces reveals that under replay conditions, the rule-based Strategy Agent heuristic predominantly reaches the identical difficulty action as the non-RL heuristic fallback.

Contradicted Prediction A10 (Guardrail Divergence). Registered prediction A10 posited that evaluating the system with canonical guardrails (B_{0c}) would alter at least one trajectory path. In replay, **zero paths differed (0/160)**. While the two implementations disagree on 11.8% of synthetic state grid points (4,728 of 39,930), this disagreement is a property of the synthetic grid rather than an observed replay or deployment frequency. The divergent grid regions correspond to extreme acoustic hesitation or low voice-confidence states; these states were not reached in the evaluated replay trajectories, where acoustic defaults remain neutral.

6.3 Evaluating Agent Specialization: Split vs. Unified Architecture (RQ2)

Trajectory Invariance under Functional Decomposition. A central motivation for multi-agent architecture is the assumption that role specialization produces different decision behavior. To illustrate this assumption, we examine Condition M_3, whose outcome is largely fixed by the architecture: The implementation exhibits a structural property implied by the architecture's authority and information topology: `CoachingPlan` contains no difficulty-control fields, `arbitrate_difficulty()` does not inspect the coaching plan, and inter-agent communication is unidirectional from Strategy to Coaching through `TurnObservation.peer`. Under this architecture, Coaching execution does not provide an information pathway back to the Strategy difficulty decision.

Table 6. Architectural and operational comparison of the Specialized Two-Agent Architecture (M_{split}) versus a Unified Monolithic Controller (M_{unified}).

Dimension	Specialized Architecture (M_{split})	Unified Architecture (M_{unified})
Agent Roles	Strategy Agent + Coaching Agent	Unified Technical-Coach Agent
State Encapsulation	Separate private ledgers (Weakness vs Coaching)	Monolithic shared session ledger
Inter-Agent Comm.	Explicit typed peer message (`TurnObservation.peer`)	Internal variable passing
Decision Equivalence	Baseline reference (M_1, 160 sessions)	$\mathbb{P} = 1.0$ (1,280/1,280 turns identical)
Serialization Micro-bench.	2 models (0.0092 ms per turn)	1 model (0.0038 ms, −58.7%)
Component Dispatch Latency	0.0440 ms per dispatch	0.0401 ms (−8.9%)
End-to-End Orchestration	+1.04 ms p50 (M_1 vs B_0 only)	Not measured
Decision-record structure	Separate per-agent decision records	Joint session decision state

Consistent with this structural property, the matched replay experiment produced identical difficulty trajectories in all 160 evaluated sessions ($M_3 \equiv M_1$, Table 3), with identical level changes (12.45), directional reversals (9.87), and score progressions (A5).

Decision Equivalence under Matched Replay. On identical observation inputs under matched heuristic rules, a standalone unified controller (`UnifiedCoachAgent`, a joint-ledger re-implementation of the same rule set) synthesizing both strategy and coaching outputs produced identical normalized decision outputs across all 160 sessions (1,280 turns):

$$\mathbb{P}(\text{Decision}_{\text{unified}} = \text{Decision}_{\text{split}}) = 1.0$$

For this comparison, a turn-level decision denotes the normalized decision tuple: Strategy difficulty action ($\{\text{Easier}, \text{Same}, \text{Harder}\}$), mode, confidence, and probe concept, along with Coaching depth, hint/example flags, tone, remediation tag, and emphasis concepts. Under matched replay, architectural decomposition did not alter observed decision outputs in this evaluation setting.

Architectural Characteristics and Operational Overhead. As synthesized in Table 6, specialization under the tested configuration establishes separate state ledgers without altering the observed decision outputs, while introducing measured operational overhead:

- *State Decoupling (Architectural Design Property):* Strategy and Coaching maintain separate memory structures, establishing separate schema boundaries for difficulty adaptation and pedagogical tracking. While this separation does not yield empirical decision divergence in replay, it maintains separate state models across roles.
- *Operational Cost (Measured Empirical Outcome):* Runtime overhead is reported at three distinct measurement scopes; only (ii) and (iii) compare M_{split} with M_{unified}: (i) *End-to-end orchestration overhead* (M_1 vs B_0; not measured for split vs unified): +1.04 ms median turn latency (B_0 0.55 ms vs M_1 1.59 ms; +2.36 ms p95; 47.7 calls/session), measuring the full orchestrator turn path and arbitration; (ii) *Component dispatch latency*: a narrower

dispatch measurement of 0.0440 ms per turn for split vs 0.0401 ms for unified (−8.9%, saving 0.0039 ms; descriptive, single-run measurements); and (iii) *Serialization micro-benchmark*: validating two Pydantic models requires 0.0092 ms vs 0.0038 ms for one model (−58.7%, saving 0.0054 ms).

7 Discussion

What Coordination Changes in Practice. Under the tested replay configuration, introducing stateful, bounded-authority agents alters the system's behavioral dynamics: M_1 exhibits fewer level changes and directional reversals than the frozen single-policy PPO baseline (estimated level-change reduction 2.67, with the cluster-bootstrap 95% interval of $[-4.50, -0.24]$ entirely below zero; reversal reduction 1.88, cluster-bootstrap 95% CI $[-3.12, -0.27]$). In the ablations, the observed trajectory changes are associated with the Strategy proposal/arbitration path: disabling Strategy (M_2) or $\tau = 0.9$ reproduces B_0, whereas removing Coaching (M_3) or peer messaging (M_6) leaves M_1 unchanged. Replay on fixed responses cannot show whether a smoother difficulty progression improves candidate learning, retention, or interview confidence; the results demonstrate algorithmic consistency under replay, not educational efficacy.

The Limits and Realities of Agent Specialization. Multi-agent frameworks motivate specialization as task differentiation [8, 17]; under the tested deterministic configuration, a joint-ledger re-implementation of the same rules produces identical per-turn decisions to the split agents ($\mathbb{P}(\text{agree}) = 1.0$ across 1,280 open-loop turns), and removing Coaching leaves difficulty trajectories unchanged ($M_3 \equiv M_1$, 160/160 sessions).

Specialization therefore functions as an architectural decomposition rather than a measured behavioral advantage:

1. *Lifecycle Decomposition (Design Characteristic):* Strategy and Coaching address distinct concerns (macro-level progression vs micro-level pedagogical presentation), providing explicit boundary contracts and independent schemas across roles.
2. *Separate State Ledgers (Architectural Property):* Separate private ledgers keep difficulty-history and concept-tracking schemas distinct without coupling domain models.
3. *Measured Operational Overhead (Empirical Outcome):* Relative to the unified agent, the split implementation shows small dispatch (0.0440 vs 0.0401 ms) and serialization (0.0092 vs 0.0038 ms) overheads. The separate +1.04 ms median end-to-end overhead (47.7 calls per session) is the cost of the multi-agent layer relative to B_0, not a split-versus-unified measurement.

This result does not establish that specialization is unnecessary in general or that the implementations remain equivalent under live, closed-loop, or LLM-backed operation; under the tested replay, authority structure, and information topology, decomposition preserved the tested decision behavior while adding measurable architectural and operational trade-offs.

Bounded Authority and Observable Coordination. Our architecture deliberately avoids open-ended negotiation or decentralized consensus: a centralized orchestrator is the system of record, and typed proposal contracts with threshold-gated arbitration make agent authority explicit and coordination observable in replay traces, without delegating state commits to autonomous components.
Implications for Agent-Native System Design. These results suggest evaluating coordination and specialization as separate architectural variables rather than one multi-agent/single-controller choice: under the tested authority and information topology, a stateful, confidence-gated agent layer measurably changed trajectory behavior, driven here by the Strategy proposal path, while splitting it into two authority-bounded roles (specialization) did not. Quantifying specialization's operational cost separately from coordination's behavioral effect, via cluster-aware replay, offers one concrete method for characterizing bounded-authority coordination before committing to a decomposition.

8 Limitations and Threats to Validity

Replay-Based Evaluation and Absence of Live Models. All empirical evaluations rely strictly on deterministic replay of 64 fixed technical-interview replay instances across eight underlying replay clusters. No generative large language model was executed during replay; the LLM gateway was stubbed as down, and follow-ups and feedback came from deterministic template fallbacks. Consequently, this paper makes **no claims** regarding coaching pedagogical quality, student learning gains, candidate satisfaction, or interview placement success. Evaluating whether multi-agent coaching improves human outcomes requires longitudinal human-participant studies.
The 8-Cluster Dependency and Sensitivity to Cluster $k = 1$. The 160 replay sessions per condition reuse the identical 8 underlying replay clusters across 20 question orders. Although cluster-bootstrap intervals (Field and Welsh [5]) account for the replay-cluster structure, the sample of 8 clusters remains limited. The sensitivity audit shows that the observed +0.59-turn increase in proactive probing occurs only in a single replay cluster ($k = 1$); in the remaining seven clusters proactive probing added zero turns. Replay dynamics cannot be assumed to generalize beyond the evaluated replay clusters. The M_1–B_0 comparison uses one frozen PPO checkpoint (`seed_123`); the B_0 oscillation is a property of that checkpoint, not of PPO in general.
Evaluator Invariance and Lack of Dynamic Candidate Adaptation. In replay mode, candidate response scores are pre-computed constants. A difficulty transition or probe therefore cannot elicit an adapted live response, which isolates coordination mechanics from human variance but precludes observing dynamic candidate adaptation.
Scope of Agent Specialization Findings. Our finding of behavioral equivalence between specialized and unified agents ($\mathbb{P}(\text{agree}) = 1.0$) applies strictly to deterministic rule- and ledger-based implementations. The split-versus-unified comparison is evaluated in an open-loop replay harness (8 turns per session;

fixed difficulty, neutral acoustic inputs, no follow-ups; agents invoked directly), and is not evidence of equivalence under live closed-loop or LLM-backed operation, where distinct prompts or LLM-backed execution may introduce behavioral variability not represented by the deterministic replay.

Heuristic Confidence Tuning and Unexercised Regions. The arbitration threshold ($\tau = 0.6$) and heuristic confidence increments ($\min(0.95, 0.5+0.1n)$) are documented engineering choices rather than mathematically optimal parameters. Because replay data lack raw acoustic waveforms (speech hesitation, acoustic pitch, voice energy), acoustic confidence was held at neutral defaults. Consequently, the grid regions (11.8% of points) where the deployed inline and canonical `rl/guardrails.py` guardrails disagree were never reached during replay, and the Strategy Agent never triggered a *stretch* proposal.

Dataset Provenance and Evaluation Scope. The 64 replay instances combine constructed (authored) answer payloads with team-generated/synthetic instances. No formal institutional ethics review was obtained. The retained replay payloads contain no names, identifiers, or sensitive personal information and serve solely as offline replay input for software evaluation.

Test Suite and Environment Bounds. Runs used standardized Python 3.12 outside historical dependency pins. The targeted coordination and specialization verification suite passed all 20 tests. The broader repository suite contains additional pre-existing failures/errors associated with documented Docker-sandbox and checkpoint/environment requirements.

9 Conclusion

Relative to the frozen single-policy baseline (B_0), introducing the specialized Strategy and Coaching agents (M_1) changes difficulty trajectories across all 160 replay sessions, with M_1 exhibiting fewer level changes (-2.67) and reversals (-1.88) alongside $+0.59$ proactive probe turns (with the additional turns occurring only in replay cluster $k = 1$). In the matched split-versus-unified comparison, M_{split} and M_{unified} produce identical per-turn decisions in the open-loop matched replay ($\mathbb{P}(\text{agree}) = 1.0$, 1,280/1,280 turns). The architectural split nevertheless establishes separate state ledgers at small measured dispatch and serialization costs (0.0440 vs 0.0401 ms; 0.0092 vs 0.0038 ms), while the multi-agent layer as a whole adds $+1.04$ ms median per-turn latency relative to B_0. Through threshold sensitivity analysis and cluster-bootstrap replay, this study provides a replay-based characterization of bounded-authority multi-agent coordination.

References

1. Alshiekh, M., Bloem, R., Ehlers, R., Könighofer, B., Niekum, S., Topcu, U.: Safe reinforcement learning via shielding. In: Proceedings of the AAAI Conference on Artificial Intelligence (2018)
2. Buşoniu, L., Babuška, R., De Schutter, B.: A comprehensive survey of multiagent reinforcement learning. IEEE Transactions on Systems, Man, and Cybernetics, Part C: Applications and Reviews **38**(2), 156–172 (2008)

3. Cemri, M., Pan, M.Z., Yang, S., Agrawal, L.A., Chopra, B., Tiwari, R., Keutzer, K., Parameswaran, A., Klein, D., Ramchandran, K., Zaharia, M., Gonzalez, J.E., Stoica, I.: Why do multi-agent LLM systems fail? arXiv preprint arXiv:2503.13657 (2025)
4. Clopper, C.J., Pearson, E.S.: The use of confidence or fiducial limits illustrated in the case of the binomial. Biometrika **26**(4), 404–413 (1934)
5. Field, C.A., Welsh, A.H.: Bootstrapping clustered data. Journal of the Royal Statistical Society: Series B (Statistical Methodology) **69**(3), 369–390 (2007). https://doi.org/10.1111/j.1467-9868.2007.00593.x
6. García, J., Fernández, F.: A comprehensive survey on safe reinforcement learning. Journal of Machine Learning Research **16**, 1437–1480 (2015)
7. Gomez, N., Batham, S.S., Volonte, M., Do, T.D.: Virtual interviewers, real results: Exploring AI-driven mock technical interviews on student readiness and confidence. In: Companion Publication of the ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW Companion '25) (2025)
8. Hong, S., Zhuge, M., Chen, J., et al.: MetaGPT: Meta programming for a multi-agent collaborative framework. In: International Conference on Learning Representations (ICLR) (2024)
9. Kim, D.K.: Artifact validity under varying agent configurations in LLM-assisted software development: A comparative analysis. Information and Software Technology **192**, 108022 (2026). https://doi.org/10.1016/j.infsof.2026.108022
10. Kim, Y., Gu, K., Park, C., Park, C., Schmidgall, S., Heydari, A.A., et al.: Capable language models can outgrow the benefits of collaboration. Nature Machine Intelligence **8**, 1157–1172 (2026). https://doi.org/10.1038/s42256-026-01268-y
11. Nygard, M.T.: Release It! Design and Deploy Production-Ready Software. Pragmatic Bookshelf (2007)
12. Raffin, A., Hill, A., Gleave, A., Kanervisto, A., Ernestus, M., Dormann, N.: Stable-Baselines3: Reliable reinforcement learning implementations. Journal of Machine Learning Research **22**(268) (2021)
13. Saltzer, J.H., Schroeder, M.D.: The protection of information in computer systems. Proceedings of the IEEE **63**(9), 1278–1308 (1975). https://doi.org/10.1109/PROC.1975.9939
14. Schulman, J., Wolski, F., Dhariwal, P., Radford, A., Klimov, O.: Proximal policy optimization algorithms. arXiv preprint arXiv:1707.06347 (2017)
15. Vaishampayan, S., Brown, C.: Will you trust me more than ChatGPT? Evaluating LLM-generated code feedback for mock technical interviews. In: Cooperative and Human Aspects of Software Engineering (CHASE) (2025)
16. Wooldridge, M., Jennings, N.R.: Intelligent agents: theory and practice. The Knowledge Engineering Review **10**(2) (1995). https://doi.org/10.1017/S0269888900008122
17. Wu, Q., Bansal, G., Zhang, J., Wu, Y., Zhang, S., Zhu, E., Li, B., Jiang, L., Zhang, X., Wang, C.: AutoGen: Enabling next-gen LLM applications via multi-agent conversation. arXiv preprint arXiv:2308.08155 (2023)
18. Yazdani, N., Mahajan, A., Ansari, A.: Zara: An LLM-based candidate interview feedback system. arXiv preprint arXiv:2507.02869 (2025)
19. Zheng, L., Chiang, W.L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E.P., Zhang, H., Gonzalez, J.E., Stoica, I.: Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. In: Advances in Neural Information Processing Systems 36 (Datasets and Benchmarks Track) (2023)